Program: **Diploma in Food Processing Technology**	
Course Code: **5429**	Course Title: **Food Chemistry II Lab**
Semester: **5**	Credits: **1.5**
Course Category: **Program Elective**	
Periods per week: **3(L:0 T:0 P:3)**	Periods per semester: **45**

Course Objectives:

- Determine the presence of various components in the given food sample
- Study the effects of enzymes in relation to different parameters
- Study the biochemical changes in various samples

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Awareness about food chemistry		Food chemistry II	V

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Apply techniques for estimating ascorbic acid in lemon juice, qualitatively estimate calcium in a given sample, and analyze water for potable and food purposes.	8	Applying
CO2	Apply knowledge of enzyme concentration and temperature effects on enzyme kinetics.	14	Applying
CO3	Determine the protein content and detect the presence of glucose and protein in a given sample using Lowry's method, Benedict test, sulfosalicylic acid test, biuret test, and heat coagulation test.	15	Applying
CO4	Apply the knowledge of protein reactions and functional properties samples in various contexts.	8	Applying

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline

Module Outcomes	Descrip tion	Duration (Hours)	Blooms Taxonomy Level
CO1	**Apply techniques for estimating ascorbic acid in lemon juice, qualitatively estimate calcium in a given sample, and analyze water for potable and food purposes.**		
M1.01	Estimation of ascorbic acid in lemon juice.	2	Understanding
M1.02	Qualitative estimation of a Calcium in given sample	3	Applying
M1.03	Analysis of water for potable and food purposes	3	Applying
CO2	**Apply knowledge of enzyme concentration and temperature effects on enzyme kinetics.**		
M2.01	Study the effect of enzyme concentration on enzyme kinetics	6	Applying
M2.02	Study the effect of enzymes on various temperatures	6	Applying
	Lab Test –I	2	
CO3	**Determine the protein content and detect the presence of glucose and protein in a given sample using Lowry's method, Benedict test, sulfosalicylic acid test, biuret test, and heat coagulation test.**		
M3.01	Determine the Protein content of a given sample by Lowry's method	3	Applying
M3.02	Check the presence of Glucose (Carbohydrate) in sample by conducting benedict test	3	Applying

M3.04	Detect the presence of protein by sulfosalicylic acid test	3	Applying
M3.03	Detect the presence of protein by biuret test	3	Applying
M3.04	Detect the presence of protein by heat coagulation test.	3	Applying
CO4	**Apply the knowledge of protein reactions and functional properties samples in various contexts.**		
M4.01	Investigate of some reactions of proteins	3	Applying
M4.02	Investigate of some functional properties of proteins	3	Applying
	Lab Test– II	2	

Text/Reference

T/R	Book Title/Author
R1	The chemical analysis of foods and food products, by Morris B. Jacobs, III Edition,CBS Publishers and distributors New Delhi.
R2	ISI hand book of food analysis
R3	Hand book of analysis and quality control for fruit and vegetable products, by S.Ranganna, II Ed., Tata McGraw Hill Publishing Co. New Delhi.
R4	Official Method of analysis of AOAC

Online Resources

Sl.No	Website Link
1.	https://info.gbiosciences.com/blog/protein-estimation-methods
2.	https://hbmahesh.weebly.com/uploads/3/4/2/2/3422804/estimation_of_protein_by_lowrys_method.pdf
3.	https://microbiologyinfo.com/benedicts-test-principle-composition-preparation-procedure-and-result-interpretation/
4.	teachmephysiology.com/biochemistry/molecules-and-signalling/enzyme-kinetics/